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Cell Rep. Author manuscript; available in PMC 2015 May 22.

Published in final edited form as:

Cell Rep. 2014 May 22; 7(4): 935–937. doi:10.1016/j.celrep.2014.05.004.

Limitations of Preimplantation Genetic Diagnosis for Mitochondrial DNA Diseases

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Steffann et al. highlight significant differences in predicting the transmission of mitochondrial DNA (mtDNA) mutations by preimplantation genetic diagnosis (PGD) in human and rhesus macaque embryos. We previously demonstrated considerable segregation of mtDNA variants between daughter blastomeres within a monkey 8-cell embryo, implying that sampling and analyzing of one or two blastomeres may not be fully predictive of total mutation load in the remaining embryo. Moreover, monkey offspring or embryonic stem cell lines produced from heteroplasmic embryos were nearly homoplasmic, suggesting that mtDNA mutation levels may increase drastically in comparison to preimplantation embryos due to a genetic bottleneck (Lee et al., 2012).

The authors reviewed several clinical studies describing mutation distribution among blastomeres in preimplantation human embryos derived from patients carrying one of several mtDNA mutations. Analysis of a minimum of two biopsied blastomeres from each cleaving human embryo demonstrated comparable mtDNA mutation levels among daughter blastomeres. Thus, the authors argue that in contrast to animal models, PGD in human embryos is a highly reliable approach to selecting embryos with sufficiently low mtDNA mutation levels for transfer.

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As an explanation for differences seen between nonhuman primate and human embryos, the authors point out that our rhesus monkey study is based on an “artificial” mixture of two mtDNA haplotypes in unfertilized oocytes, whereas heteroplasmy in human embryos occurs “naturally.” As we discussed in our study, we cannot exclude that such an extreme case of heteroplasmy involving mitochondrial sequence polymorphisms between two distant genomes could have contributed to the segregation patterns seen in monkey embryos and offspring.

However, differences in the specific molecular techniques used to measure mtDNA heteroplasmy levels in human and monkey embryo studies may also have affected the interpretation of results. In particular, our primate study utilized the amplification refractory mutation system quantitative PCR (ARMS-qPCR) assay. ARMS-qPCR is a highly sensitive quantitative assay that allows accurate assessment of heteroplasmy load at a single-cell level (Lee et al., 2012). In contrast, the human studies cited by Steffann et al. were based on semiquantitative fluorescent PCR followed by restriction enzyme digestion, which is associated with higher technical errors in heteroplasmy assessment due to heteroduplex formation or incomplete digestion (Monnot et al., 2011).

Regardless of the technical validity of mtDNA mutation-load assessments performed in preimplantation embryos, the ultimate validation of PGD success is the birth of healthy children with very low-level or undetectable mtDNA mutation load. The authors cite live births of four children who were conceived by in vitro fertilization (IVF) with embryo selection post-PGD in families carrying pathogenic mtDNA mutations. Mutation loads in newborn buccal or cord blood cells were apparently similar to the value predicted by PGD (Monnot et al., 2011; Steffann et al., 2006; Treff et al., 2012). Although very limited clinical follow-up was reported in these cases, it is on the basis of these studies that Steffann et al. conclude that PGD is a reliable technique for lowering or completely eliminating the maternal transmission of pathogenic mtDNA mutations to children.

One of these four reported PGD cases involved a family affected with mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS) syndrome due to the common m.3243A>G mutation in the mtDNA (Treff et al., 2012). After two IVF cycles and trophectoderm biopsy, a male embryo with the lowest mutation load (12% heteroplasmy) was identified and transferred, resulting in viable pregnancy. Although the authors’ report implied that the baby was healthy at birth and unlikely to manifest symptoms of MELAS later in life due to “tissue-specific mutation loads ranging from undetectable to 15%” by unspecified “clinical” molecular assay, the actual clinical details of that case were significantly more complicated. Specifically, the infant was delivered pre-term at 34 3/7 weeks’ gestation, weighing 1,890 g due to severe preeclampsia with placental atresia. The child then presented to the Mitochondrial-Genetics Clinic at The Children’s Hospital of Philadelphia with multiple medical problems that included cyanotic spell associated with hypoxemia at 1 month of age, recurrent hypoglycemia (53–64 mg/dl) that self-resolved over time, recurrent hospitalizations for infections with prolonged recovery and associated dehydration, a prolonged period of projectile vomiting associated with *C. difficile* infection, a prolonged febrile seizure, gastresophageal reflux disease, mild developmental delays (sat unassisted at 8 months, walked at 16 months, pointed at 18 months, spoke his first word at

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18 months), and behavioral problems including mild hyperactivity, difficulty calming, frequent temper tantrums, head tapping, head banging on the wall and floor, and spinning. Upon physical exams at 6 weeks and 18 months of age, he was found to have decreased pupil photoreactivity and several minor dysmorphic features, including relative macroscaphocephaly, mildly coarse facies, thickened superior ear helix, epicanthal folds, short sublingual frenulum, mild fifth finger clinodactyly, and small umbilical hernia. His brain MRI at age 15 months revealed prominence of the ventricular system and extra-axial cerebrospinal fluid; prominent Virchow-Robin spaces; pineal cyst; increased FLAIR signal involving the posterior periventricular white matter, suggestive of incomplete myelination; and multiple areas of scattered punctate signal abnormality, most extensively in the cerebellum, right temporal-occipital border, and right occipital lobe, that were indicative of prior intracranial and intraventricular hemorrhage. Extensive hematology evaluation did not reveal an underlying bleeding disorder.

Metabolic laboratory studies of his blood and urine, performed at these clinic visits, identified intermittent mild lactic acidemia and an elevated ratio of lactate to pyruvate—including an occurrence at the time of an intercurrent febrile illness (Table S1)—generalized aminoaciduria, and elevated triglycerides (245 mg/dl; normal, < 125). Multiple genetic studies were performed and did not reveal genome-wide chromosomal copy-number aberrations or identifiable nuclear gene mutations. However, ARMS-qPCR analysis performed by a CLIA-certified mitochondrial genetics diagnostics laboratory on samples collected from the boy at the ages of 6 weeks and 18 months demonstrated mutant m.3243A>G heteroplasmy loads of 47% and 46% in blood and 52% and 42% in urine, respectively (Table S1).

It is not clear whether this child's complex neurologic, developmental, and multisystem problems relate to his m.3243A>G mutation, the blastocyst trophectoderm biopsy itself, or other aspects of the IVF procedure. While some of the child's features are neither specific nor typical for MELAS, they could potentially relate to problems experienced in the perinatal course, such as preeclampsia and intrauterine growth restriction, which are known complications of the m.3243A>G mutation. Studies on the correlation between m.3243A>G mutation heteroplasmy load and clinical manifestations suggest that a 50% heteroplasmy level is on the borderline of where phenotypic expression of MELAS is expected (Jeppesen et al., 2006), and it is often not presented until later in life beyond the early newborn and childhood period. It is also well known that blood m.3243A>G heteroplasmy levels as low as 10% increase one's risk of developing diabetes mellitus and hearing loss with age. Blood m.3243A>G mutation levels are known to fall with age, but manifestation of clinical symptoms may not correlate with mutation load. Regardless, careful clinical follow-up throughout the patient's life is indicated.

This case highlights that important uncertainties remain in predicting "safe" mtDNA mutation levels sufficient to assure long-term health in children born after PGD. A phenomenon that might be relevant to his outcome is the existence of the mtDNA bottleneck that occurs during the early peri-implantation period, suggesting that mutation levels in liveborn children may change significantly compared to heteroplasmy levels determined in preimplantation embryos. As recognized above, another potential problem may relate to

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limitations of the technical accuracy of the specific molecular assays implemented for quantification of mtDNA mutation levels at a single-cell level in biopsied preimplantation embryos. There may also be PGD-related factors specific for the cell type or timing of biopsy, as the PGD case reported by Treff et al. (2012) was based on a trophectoderm biopsy while the other studies reported by Steffan et al. employed blastomere sampling.

Although the PGD heteroplasmy selection procedure based on the blastomere biopsy appeared more successful in predicting similar heteroplasmy levels in the other cases of three liveborn children cited by Steffann et al., additional limitations of this method must be considered (Wallace and Chalkia, 2013). A major concern is that PGD that simply selects for the embryo having the lowest heteroplasmy level may reduce, but is exceedingly unlikely to eliminate, the risk of transmitting mtDNA mutations. Additionally, women may not produce oocytes, and hence embryos, with levels of mutant mtDNA that are low enough to be acceptable for transfer. No exact recommendations exist for the determination of an acceptable heteroplasmy level that may be “safe” to select, because this likely depends on the mutation type, specific disease manifestations, and family history (Poulton and Bredenoord, 2010). Given the possibility of random and rapid changes in mtDNA heteroplasmy, any level of mutant mtDNA present in embryos could lead to clinical disease that presents either in childhood or later in life and is more likely to result in maternal transmission of disease in future generations. A recently reported model of mtDNA heteroplasmy inheritance predicts that transfer of an embryo having a mutation level above 5% will have a significant chance of disease recurrence (Samuels et al., 2013). Therefore, many families now request transfers of embryos with a mutation threshold of 5% or less (Sallevelt et al., 2013), although these may not be present in a given IVF cycle or individual case.

Another important PGD-related consideration is that at least two cells are removed from the 8-cell embryo, which significantly reduces its developmental potential and chance of developing into a viable pregnancy resulting in the birth of a healthy child. For example, nine IVF/PGD cycles that were initiated in four families having pathogenic mtDNA mutations resulted in just one viable pregnancy (Sallevelt et al., 2013).

Families with known pathogenic mtDNA mutations who are searching for improved reproductive options to reduce the chances of maternal disease transmission to their offspring will likely be better served by emerging methodologies that do not just select, but actively eliminate, mutant mtDNA in oocytes or embryos. In particular, spindle transfer (Lee et al., 2012; Tachibana et al., 2013) results in < 1% mutant mtDNA carryover to the resulting embryo. This reproductive technique will likely provide a more reliable and effective medical alternative to classical PGD for mtDNA disease.

Until then, the current ambiguities of PGD outcomes for families with mtDNA mutations will require careful evaluation. Further applications should be carried out as a part of clinical research trials under appropriate regulatory oversight, to ensure that parents are fully informed about the experimental nature and high risk of mtDNA mutation transmission to the offspring with that procedure. Additionally, it is critical that all children born after PGD with mtDNA heteroplasmy load selection are carefully followed long-term to monitor any

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changes in mtDNA mutation levels and associated clinical symptoms that would warrant treatment for mitochondrial disease.

Given the unpredictable nature of changes in mtDNA heteroplasmy levels that occur during development, further translational research investigations in relevant animal models are warranted. Research in rhesus macaques has provided a particularly valuable preclinical experimental system in which to better understand mechanisms of mtDNA inheritance and to develop novel treatment options for the prevention of intergenerational transmission of mtDNA diseases (Lee et al., 2012; Tachibana et al., 2013). We also highlight a need for the development of additional animal models of naturally arising pathogenic mtDNA mutations that will be essential for a better understanding of human biology and disease.

We hope that work of Steffann et al. and our response will contribute to a clearer understanding of the complexities in predicting mtDNA disease transmission and serve as a useful platform to stimulate further discussions.

Supplementary Material

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Acknowledgments

This work was supported by National Institutes of Health grants R01-HD063276, R01-HD057121, R01-HD059946, R01-EY021214, P51-OD011092 (S.M.), and R03-DK082446 (M.J.F.), as well as a grant from the Leducq Foundation (S.M.).

References

Jeppesen TD, Schwartz M, Frederiksen AL, Wibrand F, Olsen DB, Vissing J. Muscle phenotype and mutation load in 51 persons with the 3243A>G mitochondrial DNA mutation. Arch Neurol. 2006; 63:1701–1706. [PubMed: 17172609]

Lee HS, Ma H, Juanes RC, Tachibana M, Sparman M, Woodward J, Ramsey C, Xu J, Kang EJ, Amato P, et al. Rapid mitochondrial DNA segregation in primate preimplantation embryos precedes somatic and germline bottleneck. Cell Rep. 2012; 1:506–515. [PubMed: 22701816]

Monnot S, Gigarel N, Samuels DC, Burlet P, Hesters L, Frydman N, Frydman R, Kerbrat V, Funalot B, Martinovic J, et al. Segregation of mtDNA throughout human embryofetal development: m. 3243A>G as a model system. Hum Mutat. 2011; 32:116–125. [PubMed: 21120938]

Poulton, J.; Bredenoord, AL. Neuromuscul Disord; 174th ENMC international workshop: Applying pre-implantation genetic diagnosis to mtDNA diseases: Implications of scientific advances; 19–21 March, 2010; Naarden, The Netherlands. 2010. p. 559-563.

Sallevelt SC, Dreesen JC, Drüsedau M, Spierts S, Coonen E, van Tienen FH, van Golde RJ, de Coo IF, Geraedts JP, de Die-Smulders CE, Smeets HJ. Preimplantation genetic diagnosis in mitochondrial DNA disorders: challenge and success. J Med Genet. 2013; 50:125–132. [PubMed: 23339111]

Samuels DC, Wonnapinij P, Chinnery PF. Preventing the transmission of pathogenic mitochondrial DNA mutations: Can we achieve long-term benefits from germ-line gene transfer? Hum Reprod. 2013; 28:554–559. [PubMed: 23297368]

Steffann J, Frydman N, Gigarel N, Burlet P, Ray PF, Fanchin R, Feyereisen E, Kerbrat V, Tachdjian G, Bonnefont JP, et al. Analysis of mtDNA variant segregation during early human embryonic development: a tool for successful NARP preimplantation diagnosis. J Med Genet. 2006; 43:244–247. [PubMed: 16155197]

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Tachibana M, Amato P, Sparman M, Woodward J, Sanchis DM, Ma H, Gutierrez NM, Tippner-Hedges R, Kang E, Lee HS, et al. Towards germline gene therapy of inherited mitochondrial diseases. Nature. 2013; 493:627–631. [PubMed: 23103867]

Treff NR, Campos J, Tao X, Levy B, Ferry KM, Scott RT Jr. Blastocyst preimplantation genetic diagnosis (PGD) of a mitochondrial DNA disorder. Fertil Steril. 2012; 98:1236–1240. [PubMed: 22921075]

Wallace DC, Chalkia D. Mitochondrial DNA genetics and the heteroplasmy conundrum in evolution and disease. Cold Spring Harb Perspect Biol. 2013; 5:a021220. [PubMed: 24186072]